

Solution Requirements

Date	1 July 2026
Team ID	xxxxxx
Project Name	HouseRent – House Rental Management System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should *do*).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description
1	Authentication	Users shall securely register and log in using email and password with JWT-based authentication.
2	Authorization Levels	The system shall support three roles: Admin, Property Owner, and Tenant, each with specific permissions.

3	External Interfaces	The system shall connect with MongoDB for data storage and use RESTful APIs for communication between frontend and backend.
4	Transaction Processing	The system shall process property bookings, manage booking requests, and securely exchange data between users and the database.
5	Reporting	The system shall provide dashboards for property owners and administrators to view property listings and booking information.
6	Business Rules	Only authenticated users can access protected features. Property owners can manage only their own listings, while tenants can book available properties.
7	Compliance to Laws or Regulations	The system shall protect user data by following secure authentication practices and basic data privacy principles.
8	Other	The system shall allow users to search, filter, view, add, update, delete, and book rental properties.

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metrics / Acceptance Criteria
1	Performance & Speed	The application should provide fast response times for user requests and database operations.	Page loads in seconds

2	Scalability	The system should support increasing numbers of users and property listings without significant performance degradation.	Supports u registered u
3	Security & Data Privacy	Passwords shall be encrypted using bcrypt, and authentication shall be secured using JWT.	Secure aut encrypted p
4	Reliability & Availability	The application should remain available with minimal downtime and recover gracefully from unexpected failures.	99% system
5	Usability & Accessibility	The application should provide a responsive, intuitive interface that works across desktops, tablets, and mobile devices.	Responsive navigation
6	Maintainability & Portability	The application should use a modular MVC architecture for easy maintenance and deployment across different environments.	Modular co architecture